

SAVONLINNA, Finland

WMO: 029480

Lat: 61.946N

Lon: 28.932E

Elev: 95

StdP: 100.19

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -27.8	(c) -24.0	(d) -30.8	(e) 0.2	(f) -27.6	(g) -26.7	(h) 0.3	(i) -24.1	(j) 8.0	(k) -5.0	(l) 7.2	(m) -4.3	(n) 0.8	(o) 270	(p) 0.624

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	26.9	18.7	25.0	17.4	23.1	16.5	19.8	24.4	18.7	23.0	17.6	21.7	3.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.3	21.5	17.0	12.3	20.6	16.0	11.5	19.4	57.1	24.7	53.3	22.8	50.0	22.0	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.4	6.5	5.9	-30.5	28.8	3.9	2.4	-33.3	30.5	-35.6	31.9	-37.7	33.2	-40.6	35.0
			-30.6	21.2	3.8	1.5	-33.4	22.3	-35.6	23.1	-37.8	24.0	-40.6	25.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.9	-8.4	-8.1	-3.9	2.9	9.2	13.7	16.7	15.0	10.2	4.2	-0.8	-4.6
	DBStd	9.98	7.55	7.25	5.28	3.63	4.44	3.73	3.28	3.23	3.34	3.86	4.55	6.47
	HDD10.0	2801	571	508	431	215	68	9	1	3	37	183	323	452
	HDD18.3	5299	829	741	689	463	282	147	72	111	243	439	573	711
	CDD10.0	579	0	0	0	2	45	119	208	158	44	2	0	0
	CDD18.3	36	0	0	0	0	1	7	21	8	0	0	0	0
	CDH23.3	348	0	0	0	0	22	74	184	67	0	0	0	0
	CDH26.7	57	0	0	0	0	1	11	37	7	0	0	0	0
Wind	WSAvg	2.8	2.8	2.7	2.9	2.7	2.9	2.9	2.6	2.5	2.7	2.9	3.1	3.2
Precipitation	PrecAvg	627	49	37	34	28	42	66	71	71	57	61	57	51
	PrecMax	768	76	73	67	73	131	108	145	133	141	134	114	87
	PrecMin	466	18	5	8	6	9	23	21	17	19	15	9	22
	PrecStd	69	17	19	15	15	27	21	30	34	30	27	24	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.2	4.2	9.6	18.0	26.1	28.1	29.8	27.9	20.8	13.5	9.0	6.9
		MCWB	2.5	2.5	4.4	10.5	15.4	18.1	20.5	19.5	15.7	11.6	8.1	6.1
	2%	DB	2.2	2.4	6.0	14.4	23.1	25.5	27.6	25.4	18.6	11.9	7.4	4.8
		MCWB	1.7	1.6	2.7	8.6	14.0	17.1	19.5	18.0	14.8	10.5	6.8	4.1
	5%	DB	1.2	1.4	3.9	11.9	20.3	23.2	25.5	23.5	17.1	10.9	6.3	3.2
		MCWB	0.8	0.9	1.6	6.6	12.7	15.9	18.2	17.2	13.9	9.8	5.7	2.6
	10%	DB	0.3	0.6	2.4	9.6	17.9	21.0	23.4	21.6	15.6	9.9	5.1	2.1
		MCWB	0.0	0.3	0.8	5.4	11.2	14.7	17.2	16.5	13.3	9.0	4.6	1.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	2.8	4.9	12.1	17.2	20.1	21.9	21.1	16.7	12.1	8.3	5.9
		MCDB	2.9	3.5	8.4	16.0	23.4	25.5	27.3	25.8	18.6	13.1	8.8	6.4
	2%	WB	1.7	1.8	3.0	9.5	15.3	18.3	20.4	19.1	15.5	11.0	6.9	4.0
		MCDB	2.0	2.4	5.2	13.2	20.4	23.5	25.5	23.1	17.7	11.7	7.2	4.4
	5%	WB	0.9	1.0	2.0	7.4	13.6	16.9	19.3	18.1	14.5	10.0	5.7	2.7
		MCDB	1.1	1.3	3.2	11.0	19.2	21.4	23.7	22.3	16.5	10.7	6.2	3.1
	10%	WB	0.1	0.3	1.1	5.6	12.0	15.6	18.1	17.1	13.6	8.9	4.6	1.7
		MCDB	0.3	0.6	2.1	8.8	16.8	20.0	22.0	20.7	15.3	9.6	5.1	2.1
Mean Daily Temperature Range	5% DB	MDBR	5.8	6.5	7.9	8.9	10.5	9.6	9.3	8.8	7.1	4.8	3.9	4.7
		MCDBR	3.9	4.7	8.8	13.4	14.7	13.1	12.4	12.2	9.5	5.5	4.1	3.6
		MCWBR	3.9	4.3	6.1	7.8	7.0	5.9	5.2	5.8	6.2	4.3	4.0	3.5
	5% WB	MCDBR	4.0	4.3	7.0	11.9	12.2	11.2	10.9	10.4	8.3	4.9	4.1	3.4
		MCWBR	4.0	4.0	5.3	7.4	6.5	5.6	5.3	5.5	5.7	4.3	4.1	3.4
Clear-Sky Solar Irradiance	taub	0.243	0.291	0.318	0.358	0.354	0.350	0.375	0.359	0.336	0.317	0.261	N/A	
	taud	2.034	2.117	2.195	2.340	2.446	2.490	2.435	2.461	2.514	2.403	2.122	N/A	
	Ebn at Noon	430	653	783	830	867	877	841	824	779	643	405	N/A	
	Edh at Noon	37	70	94	101	98	96	99	89	71	55	34	N/A	
All-Sky Solar Radiation	RadAvg	0.22	0.78	2.27	3.65	4.92	5.30	5.11	3.92	2.22	0.88	0.23	0.11	
	RadStd	0.03	0.19	0.32	0.42	0.58	0.32	0.48	0.47	0.25	0.13	0.04	0.01	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)		+0.54	N/A	N/A	N/A	N/A	N/A	-179	N/A	N/A	N/A

Nomenclature: See separate page